

Effects of land use and forest management on soil carbon in the ecoregions of Maryland and adjacent eastern United States



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ABSTRACT

The impacts of forest-related land use and management on soil organic carbon (SOC) stocks have been investigated through years of primary research and review articles. This attention is justified given the importance of land use and management to greenhouse gas mitigation, soil and forest productivity, and other ecosystem services. However, there is a gap of scale and scope between site-level studies that control for sources of variation, producing high-confidence results for limited locations, and the broad reviews that offer more general conclusions. The present analysis is intended to fill that gap. Here, we focus on six ecoregions of the eastern United States, and integrate meta-analysis of published literature with synthesis of geo-referenced soil observations to: (1) test whether common land use and management practices quantitatively impact SOC; (2) identify key sources of variation in these effects; (3) assess how sources of variation translate to decisions about land use and management at ecoregional to landscape levels. Results corroborate general trends, such as O horizon SOC losses with harvesting and fire and SOC gains during reforestation, but provide greater detail about the influence of specific practices and site-level controls on SOC stocks and change in the study region. Results also show that: (1) harvest impacts depend upon landform and soil taxonomy; (2) harvesting forests that are recovering on previously cultivated lands decreases SOC; (3) tree biomass and SOC recovery increase concurrently during reforestation; (4) specific harvest, site preparation, and fire management practices affect the magnitude and variability of changes in SOC. Perhaps more importantly, ecoregional classification and soil taxonomy provide spatial frameworks for placing quantitative estimates of SOC stocks and changes in the geographic context of the study region, providing greater detail and specificity for individuals and institutions concerned with SOC management at more localized levels.

1. Introduction

Individuals and institutions concerned with land and soil management have long known that soil organic matter (SOM), which is primarily comprised of soil organic carbon (SOC), is critical to agricultural and forest productivity and myriad other ecosystem services (Vance, 2000). More broadly, the central roles played by soils in greenhouse gas mitigation are acknowledged in greenhouse gas reporting and policy (Domke et al., 2017; Griscom et al., 2017; Minasny et al., 2017). Unfortunately, there is a wide gap in scale and scope between such high-level national and global reviews and the SOC assessments needed to inform decision making at sub-regional, landscape, and project levels. Broad reviews provide general answers to critical questions, such as the distribution of SOC stocks and their sensitivity to management at large scales (Jobbagy and Jackson, 2000; Ogle et al., 2005; Nave et al., 2010; Scharlemann et al., 2014). However, land owners, forest managers, policy and reporting professionals often need information for specific locations, where generalizations (Achat et al., 2015) frequently break

down (Clarke et al., 2015; Vance et al., 2018). In such cases, targeted synthesis of empirical data provides a way to assess SOC management under the geographic, land use, and management constraints present in the regions, landscapes, and projects where decisions are implemented.

The ecoregions of Maryland, which extend into adjacent states from the Mid-Atlantic down to the Southern Appalachians, are home to some of the most biologically diverse forests, wide-ranging soils, and complex physiography in the U.S.A. (Butler et al., 2015; Butler-Leopold et al., 2018). Complex topography at ecoregional to landscape levels drives corresponding variation in climate, vegetation, and soil. These sources of geographic and ecologic variation interact with a large and patchily distributed population, such that its history of land use, abuse, and management is also a history of soil change. Through agriculture and fire, Native Americans impacted the soils and ecosystems of the Central Appalachians and Mid-Atlantic for at least centuries before Euro-American colonization (Fesenmyer and Christensen, 2010; Springer et al., 2010). During the centuries following Euro-American colonization, more widespread deforestation, cultivation, and a lack of soil

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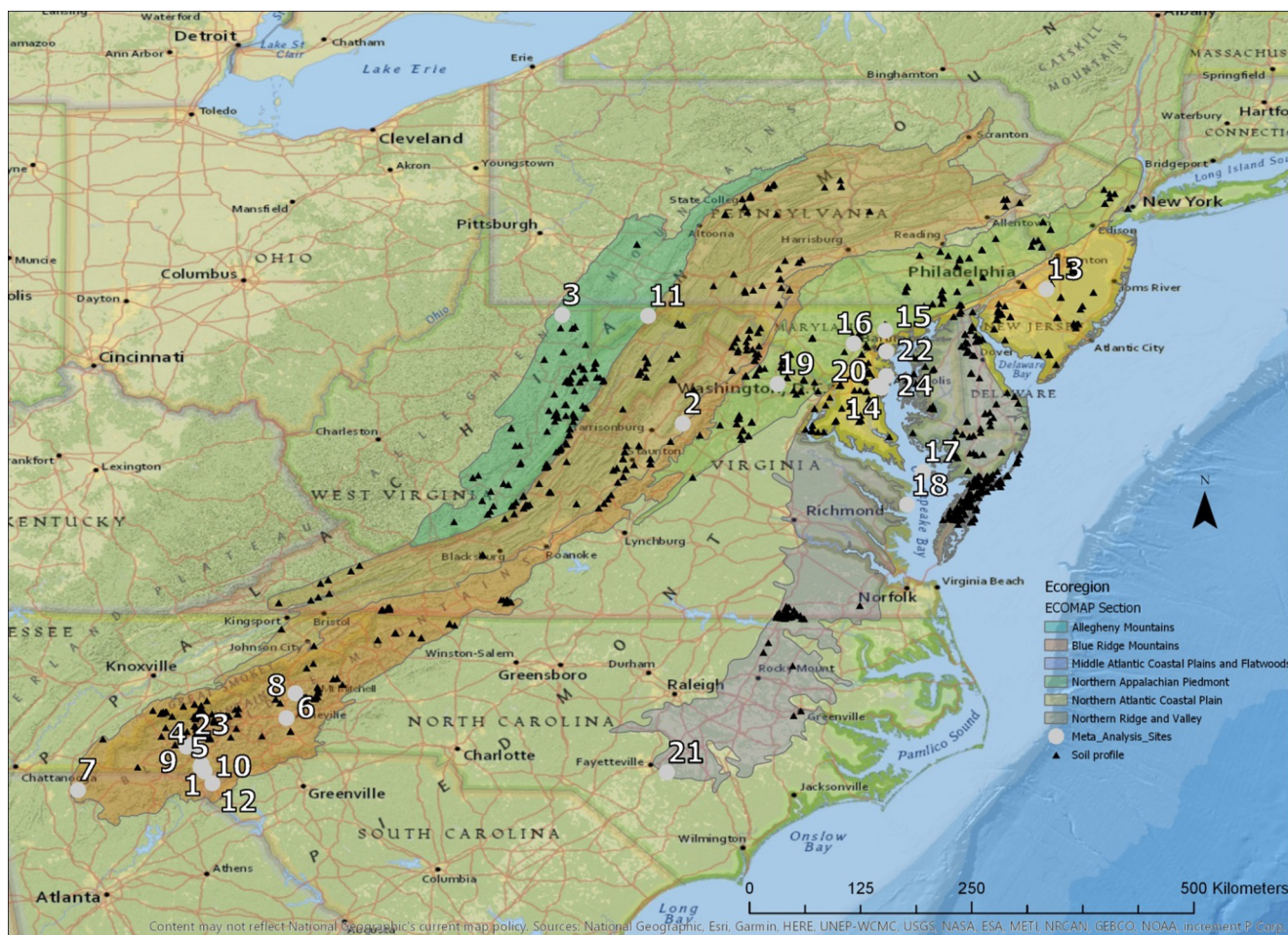


Fig. 1. Map of study area showing ecoregional units (ECOMAP sections), sites of published studies utilized in meta-analysis (large gray circles), and soil observations from the ISCND Database (small black triangles). Numbers associated with meta-analysis study sites correspond to the publications listed in Table 1.

conservation practices drove erosion, compaction, and SOC depletion (Conant et al., 2004; Geleta et al., 2014). In the late 1800s and early 1900s, widespread forest harvesting and fires changed forests and soils, most notably and fundamentally through conversion of high-elevation coniferous Spodosol ecosystems into hardwood-dominated forests with Inceptisols (Nauman et al., 2015a). Changes in the stocks and vertical distribution of SOC accompanied these changes. Mining, especially of coal, has also significantly transformed soils and forest ecosystems of the Appalachian Mountains, converting (through reclamation) many forest ecosystems into grasslands with lower SOC stocks (Simmons et al., 2008). Altogether, the long history of land use in this region has impacted its soils and their C stocks, from subtle and localized fingerprints of subsistence land uses, to severe degradation and soil order-level changes. Interest and effort have been increasing for years towards using management to restore soils and ecosystems to states and variability similar to those before widespread deforestation, agriculture, and mining (Avera et al., 2015; Matlack, 2013; Nauman et al., 2015b). Restoration goals raise important questions, however. What are appropriate targets and metrics for assessing success? How will continued changes in climate and other pressures influence the feasibility of management alternatives, and their impacts on soils? Comprehensive answers to these questions must address SOC. Prior information or geographically constrained predictions are often unavailable, but focused synthesis of empirical data can help set expectations in regions where historical complexity, geographic variation, and ongoing change challenge generalizations from the large-scale literature.

Empirical analyses of land use and management impacts on SOC have their own challenges and limitations, from study design, to field

methods execution, to questions of inferential stability under conditions or over timescales not represented in the data (Hurlbert, 1984; Throop et al., 2012; Yanai et al., 2000). Thus, empirical tests of soil observations gain strength through the application of multiple independent or complementary approaches. In this analysis, we combine approaches previously applied at larger spatial scales (biomes to major ecoregions) to quantify the effects of forest-related land use and management on SOC (e.g., Nave et al., 2010, 2013, 2018). First, we use meta-analysis of published literature, synthesizing controlled studies to hold extraneous factors constant and address specific land use and management treatments of interest. Meta-analysis of published studies produces high-confidence inferences, but these are based on the limited number of sites available in the literature. We allay this limitation by synthesizing soil observational data across the region of interest. Analysis of observational data offers less ability to control for extraneous sources of variation than meta-analysis of literature data; thus the effects of management must be strong enough to detect even through the influences of other, uncontrolled sources of variation. Failing to detect impacts in observational data that are challenging to detect even in a controlled study (Homann et al., 2001) is to be expected. Corroborating a meta-analysis trend with observational data suggests the trend is robust; finding a fundamentally different pattern reveals site-dependence and limitation of generalities based on meta-analysis. In the present work, we harness these approaches to address three objectives for the ecoregions of Maryland and adjacent states: (1) test whether common land use and management practices quantitatively impact SOC; (2) identify key sources of variation in these effects; (3) assess how these sources of variation, as contextualized by published studies, translate to

decisions about land use and management at ecoregional to landscape levels.

2. Methods

2.1. Study area

We used the U.S. Department of Agriculture – Forest Service ECOMAP classification system (Cleland et al., 1997; McNab et al., 2007), in similar fashion to previously published work (Nave et al., 2018; 2019), to identify the ecoregional sections present in Maryland (Fig. 1) and synthesize data across their full extent. These 6 ecoregional sections tier beneath the province level of ECOMAP's hierarchy of ecological units, and are defined by their unique, but internally consistent climate, landforms, parent materials, vegetation and soil groups. Detailed descriptions of the sections (McNab et al., 2007) are briefly summarized here. Three of the sections in the study area are mountainous; these are the: (1) Northern Ridge and Valley, (2) Allegheny Mountains, and (3) Blue Ridge Mountains. These 3 sections differ in their elevation, parent material, and physiography, but share a similar climate (moist temperate, with cool summers and short, mild winters) and similar forest vegetation (mixed mesophytic and xerophytic deciduous broadleaved forests at low- to mid-elevations, giving way to conifers and shrubs at high elevations). Ice and intense rain storms associated with remnants of Atlantic hurricanes are key disturbances, and lightning-caused fires are uncommon. Soils are predominantly Inceptisols and Ultisols, formed in colluvium or residuum of acid metamorphic to calcareous sedimentary bedrock types. The 4th section in the region of interest is the Northern Appalachian Piedmont, located along the northeastern flank of the 3 mountainous sections, and is a maturely dissected peneplain that slopes to the coastal sections. Unlike the mountainous sections, where precipitation is uniformly high throughout the year, precipitation in the Northern Appalachian Piedmont is higher during its warmer summers. Vegetation consists of xerophytic and mesophytic mixtures of *Quercus* spp., *Carya* spp., and *Pinus* spp. Soils are Ultisols, Inceptisols, and Alfisols, formed in the residuum or colluvium of sedimentary or metamorphic bedrock types. The final two (coastal) ecoregional sections share a humid maritime climate with warm summers, mild winters, and abundant precipitation. The Northern Atlantic Coastal Plain is a level, weakly dissected alluvial plain, where forests are dominated by *Quercus* spp., *Carya* spp., and *Pinus* spp., and soils are Ultisols and Entisols formed in marine shales and recent sands, respectively. The Middle Atlantic Coastal Plains and Flatwoods section is weakly to moderately dissected, supports forests of *Pinus* spp., *Quercus* spp., *Taxodium* spp., and *Nyssa* spp.; soils are Ultisols and Entisols, with scattered Histosols.

2.2. Data synthesis

We synthesized published data from 24 papers identified through literature review (Table 1), and geo-located soil observations from the International Soil Carbon Network (ISCN) Database. Our literature review followed previously published methods (e.g., Nave et al., 2010). In brief, we searched for relevant peer-reviewed literature using keyword searches and reference checks using ISI Web of Science. Keyword search phrases were combinations of [Geographic Term] + [Treatment] + Soil Carbon, where [Geographic Term] was Maryland or one of the 6 ecoregional sections, and [Treatment] terms were: forest management, timber, fire, afforestation, reforestation, reclamation, restoration, soil amendments, development, and site preparation. We limited our searches to publications from 2008 to 2018, in order to add new papers to those that we included in previous studies (Nave et al., 2009; 2010; 2011; 2013). Our keyword searches returned 112 papers; of these, we included those that: (1) reported control and treatment values for soil C stocks or concentrations, (2) provided adequate metadata for use as potential predictor variables in meta-analysis, (3) presented novel

response data not included in previous studies, and (4) had at least one study site located within one of the 6 ecoregional sections. Eight publications met these criteria, in addition to 10 earlier (pre-2008) publications identified in our previous studies. We checked the references for each of these 18 publications to search for additional relevant studies, yielding 594 further papers, of which 6 met our inclusion criteria. Overall, we reviewed 725 studies and found 24 suitable for our meta-analysis (Table 1).

We assembled a dataset of geo-located soil observations using the ISCN Database Version 3.0 (Nave et al., 2017) as a starting point. Similar to Nave et al. (2018, 2019), we only used soils observed after 1 January 1989, and for each of these we extracted geo-located attributes from several concurrent remote sensing data products. These included all of the were the National Land Cover Database (NLCD) products from 1996 through 2011 (Vogelmann et al., 2001; Homer et al., 2004; Fry et al., 2011; Homer et al., 2015), and the GAP/LANDFIRE National Terrestrial Ecosystems 2011 dataset (USGS Gap Analysis Program, 2016). We extracted aboveground biomass from the National Biomass and Carbon Dataset for the Year 2000 NBCD2000 (Kellndorfer et al., 2013). We also extracted point-specific climate variables (mean annual temperature [MAT] and mean annual precipitation [MAP]) from PRISM's United States Annual Precipitation and Mean Temperature datasets (PRISM Climate Group 2015a; 2015b), and extracted and derived terrain attributes (elevation, slope, aspect) from the National Elevation Dataset (USGS, 2016). From these derived terrain attributes, we calculated a topographic landform index for each soil observation point according to the methods of McNab (1993). This index classifies the relative landscape position of each soil observation point as ridge, slope, or cove based on the visual angle from plot center to the topographic horizon in the eight principal compass directions. Because field measurements for each point were not available, we used ArcMap 10.6 and the National Elevation Dataset (USGS, 2016) to model these visual angles and calculate the topographic landform index.

2.3. Meta-analysis

The primary purpose of meta-analysis was to utilize the strength of published literature and control:treatment comparisons to assess the effects of land use and management on SOC, for the limited number of locations where such studies have been conducted with some degree of experimental rigor. Secondarily, we used meta-analysis to identify key sources of variation in treatment effects, which in turn guided the statistical approaches that we utilized in tests of the more extensive observational data. Our meta-analysis used the unweighted effect-size approach (Nave et al., 2010). Namely, we extracted control and treatment SOC values, and used these to calculate effect sizes (as the ln-transformed response ratio R), from the 24 publications described in Section 2.2 and Table 1. We then used MetaWin software (Sinauer Associates, Sunderland MA, USA) to estimate effect sizes and bootstrapped 95% confidence intervals (Hedges et al., 1999). As in past analyses, unweighted meta-analysis was necessary due to unreported or ambiguous sample sizes and variances in the papers that otherwise met our inclusion criteria. Broad treatments of interest included forest harvesting (and associated post-harvest practices), fire management (wildfire and two types of prescribed fire), and land use change (paired comparisons of natural forests or wetlands to other land uses, e.g., reforestation, wetland restoration, developed lands). Published papers reported two types of soil organic contents: SOM, measured by loss on ignition (LOI), and SOC, measured using elemental analyzers. Of the $k = 109$ response ratios we calculated, 41 were measured as SOM; we assumed for all of these that 50% of the lost mass was organic C and multiplied each LOI value by 0.5 to estimate SOC concentration (Nave et al., 2013; Pribyl, 2010). Publications also differed in units of reporting of soil organic contents; $k = 31$ (of the 109 total) reported SOC as a concentration (e.g., g kg^{-1}) rather than as the SOC stock of interest to our analyses. When SOC concentrations were accompanied by bulk

Table 1

Summary of studies used in meta-analysis (cf. Fig. 1). SOM/SOC refers to soil organic contents reported as organic matter vs. organic C; units distinguish concentrations vs. stocks. Treatments are harvest (HARV), fire (FIRE), and land use change (LUC).

Study #	Reference	SOM/SOC	Units	Treat	Sampling design
1	Mattson and Wt (1989)	SOM	Stock	HARV	O, A, E, B horizons from 3 watersheds (2 cut, 1 control) at Coweeta Hydrol. Lab.
2	Groeschl et al. (1991)	SOC	Stock	FIRE	O, A horizons from unburned, low, and high-severity areas (3 each) across 350 ha
3	Mattson and Smith (1993)	SOM	Stock	HARV	O, A horizons from 15 paired harvest/control sites in a 3000 ha research forest
4	Clinton et al. (1996)	SOC	Stock	FIRE	Pre- and post-treatment O horizons from 3 fell-and-burn sites, Nantahala N.F.
5	Knoepp and Swank (1997)	SOC	Conc.	HARV	Pre- and post-cut A horizons from 2 harvested watersheds at Coweeta Hydrol. Lab
6	Neher et al. (2003)	SOM	Conc.	HARV	A horizons from 4 control, 2 harvested, and 2 farmed sites in two NC counties
7	Hubbard et al. (2004)	SOC	Stock	FIRE	Pre- and post-treatment O horizons from 4 fell-and-burn watersheds in GA and TN
8	Fraterrigo et al. (2005)	SOC	Stock	HARV	A horizons from 6 cut and 4 control watersheds in two NC counties
9	Elliott and Knoepp (2005)	SOC	Stock	HARV	Pre- and post-cut A horizons for 9 sites across 1000 ha on the Nantahala N.F.
10	Elliott et al. (2004)	SOC	Stock	FIRE	Pre- and post-fire O horizons for a 5 ha prescribed burn on the Nantahala N.F.
11	Simmons et al. (2008)	SOC	Stock	HARV	O, A horizons from 1 control and 1 harvested watershed in western MD
12	Knoepp et al. (2009)	SOC	Stock	FIRE	Pre- and post-fire O horizons for 3 prescribed burn watersheds across NC and GA
13	Bedison et al. (2013)	SOC	Stock	LUC	OA horizons from 20 forest and 9 lawn sites on the PA and NJ Coastal Plain
14	Yesilonis et al. (2016a)	SOC	Conc.	LUC	O, A, E, B horizons from 8 old and 5 young forests at Smithsonian Env. Res. Ctr.
15	Yesilonis et al. (2016b)	SOM	Conc.	LUC	A horizons from 1 urban forest and 32 lawns in a Baltimore neighborhood
16	Waters et al. (2014)	SOM	Conc.	LUC	A horizons from 3 paired forest/lawn sites along each of 4 urban streams
17	Hunt et al. (2014)	SOC	Conc.	LUC	OA horizons from 14 farmed, 14 restored and 11 natural wetlands in DE, MD, VA
18	Fenstermacher et al. (2016)	SOC	Stock	LUC	1 m profiles from 16 farmed, 18 restored, and 14 natural wetlands in DE, MD, VA
19	Ahn and Peralta (2012)	SOC	Conc.	LUC	OA horizons from 4 restored and 2 natural wetlands in VA Piedmont
20	Ma et al. (2013)	SOC	Stock	LUC	O, A, E, B horizons from 5 old and 3 young forests at Smithsonian Env. Res. Ctr.
21	Bruland et al. (2003)	SOC	Stock	LUC	1 m profiles from 1 farmed, 1 restored, and 1 natural wetland in NC
22	Raciti et al. (2011)	SOC	Stock	LUC	A, B, C horizons from 32 lawns and 8 forests in Baltimore
23	Knoepp et al. (2004)	SOC	Conc.	FIRE	A horizons from 3 paired fell-and-burn/control sites on the Nantahala N.F.
24	Pouyat et al. (2009)	SOC	Stock	LUC	AEB horizons, 1 m profiles from 1 urban forest and 7 lawns in Baltimore

density (Db) data, we calculated SOC storage directly as the product of C concentration (%), Db (g cm^{-3}), and the thickness of the horizon or sampling layer (cm), and applied expansion factors to express SOC storage in Mg ha^{-1} . However, SOC concentrations sometimes were not accompanied by Db data. Therefore, after synthesizing the meta-analysis dataset, we derived prediction equations to calculate Db from %C using the geo-located soil observations dataset (see Section 2.4) and then performed the SOC stock calculation using the measured %C and layer thickness and predicted Db values.

We extracted potential predictor variables from each publication to address our objectives of identifying factors that predict variation in SOC responses to land use or management. We looked up missing information in other publications from the same sites, or used information about the soil series reported from those study sites from the web-based interface for the USDA-Natural Resources Conservation Service (USDA-NRCS) Official Soil Series Descriptions (<https://soilseries.sc.egov.usda.gov/osdname.aspx>). Because studies were inconsistent in their sampling approaches and site descriptions, it was necessary to create categories for many factors, in order to parse variation within and between studies into sufficiently replicated groups for meta-analysis. The complete list of factors extracted from, or assigned to, the published studies is available in Table S1. Two factors emerged as particularly important during meta-analyses: sampling depth and landform. We accommodated data across a range of sampling strategies by recording the horizon or sampling increment (as depth range in cm) for each SOC value. Then, we categorized O horizons (Oe, Oa, combined O) as a single O horizon category, and mineral soils into four different categories corresponding to the uppermost, second, third, or fourth sampled mineral soil horizon or increment. Overall, 29 of the $k = 109$ response ratios were for O horizons and 60 were for the uppermost mineral soil layer (A horizon or depth increment). The average depth of these uppermost mineral soil layers was 10.1 cm, corresponding very closely with the 10 cm depth cutoff that we later utilized in analyses of geo-located soil observations (Section 2.4). In both cases (the published literature data and the geo-located soil observations), samples of the uppermost 10 cm of mineral soil were almost always A horizons; thus, we refer to the uppermost mineral soil layer as the A horizon. Regarding landforms, fire and harvest studies consistently provided information that allowed us to place soil observations into landform groups that we

categorized as: (1) landforms of combined ridge and slope setting, including uplands and adjacent slopes; (2) sloping landforms that did not include adjacent uplands; (3) cove- or watershed-level studies in which sampling was conducted for an entire small catchment or cove landform. These groups correspond very closely to the landforms categorized for the Central and Southern Appalachians by McNab (1993), which was the source for methods we used to derive topographic landform indices for the observational data. Land use change studies were highly variable in their landform terminology, and did not provide enough information to consistently categorize landforms. Overall, our efforts to obtain predictor variables and categorize studies were more comprehensive than in past analyses, but our use of these variables was otherwise the same. Namely, we ran meta-analysis first on the full dataset ($k = 109$ response ratios) to identify the strongest overall predictor of variation, in terms of within-group (Q_B) divided by between-group heterogeneity (Q_T), then split the dataset into the groups defined by that predictor (land use/management treatment). Subsequently, we ran meta-analysis within each land use/management treatment to probe more specific sources of variation. Although it cannot test for interactions between predictor variables, this iterative approach is an effective way to address sources of variation in forest ecology and management meta-analyses, which are often based on papers that do not report the necessary information (sample sizes and variances) to conduct parametric or weighted meta-analyses (e.g., Nave et al., 2010; Wan et al., 2001).

2.4. Analysis of geo-located soil observations

Our approach to synthesizing soil observations, harmonizing them with remote sensing and other GIS-based information, and conducting statistical tests was guided by the meta-analysis phase of this study, with the overall goal of validating meta-analysis results where possible, and acquiring more detailed insights where the more extensive observational data allowed them. To conduct synthesis of observational data, we began with the same datasets, and used the same gap-filling and SOC stock computations that we have employed in other studies (Nave et al., 2018, 2019). Briefly, we began with $n = 915$ geo-located soil profiles from the study region, extracted point values from the spatial data products described in Section 2.2, and calculated SOC

stocks for the constituent horizons. For the spatially explicit attributes, we assumed that land cover attributes for soils sampled between 1 January 1989 and 31 December 1996 were represented by NLCD1992; soils from 1997 to 2001 were represented by NLCD2001; soils from 2002 to 2006 by NLCD2006; soils from 2007 to most recent (2014) by NLCD2011. We only extracted aboveground biomass data for soils matched to NLCD2001 and NLCD2006 products, and we only extracted the 2011 GAP/LANDFIRE land cover attributes for soils matched to the NLCD2011 product.

For statistical tests comparing SOC stocks of soils differing in land use, we made a series of groupings in order to: (1) place soil observation locations into broad NLCD land cover groups, and then (2) infer specific information about management at those locations based upon soil profile descriptions (and in some cases their GAP/LANDFIRE attributes). Land cover groups were as follows: soil observation locations designated as “cultivated” by NLCD were retained in a cultivated group, locations identified as grassland or pasture/hay were grouped as “forage,” and all NLCD classes under the “developed” category were grouped as such. Shrub/scrub locations were considered to be forestland, and all three forest compositional groups used by NLCD (deciduous, evergreen, mixed) were grouped into the “forest” category. Soil observations with NLCD classes of barren land or open water were grouped into a “barren/water” category. Woody and emergent herbaceous wetlands were grouped into a “wetland” category.

Next, within forests, we assumed that locations classified as shrub/scrub represented recently harvested forests, unless possessing Ap horizons (plow layers; see Nave et al., 2018), in which case they were assumed to represent recently harvested forests growing on formerly cultivated lands, a condition we refer to as “reforested + harvested.” For forests classified by NLCD as forest cover types, we termed those possessing Ap horizons as “reforestation” (on formerly cultivated land), and those lacking Ap horizons as “natural forest,” a term selected for the group to distinguish it from reforestation, harvested forests, or reforested + harvested forests. In several cases, GAP/LANDFIRE attributes (which are more specific than NLCD) flagged a location as a harvested forest or plantation even when it was assigned to the grassland or wetland NLCD categories. For these, as well as all forests that were taken to represent a harvested condition, we inspected recent satellite imagery for evidence of recent management at that location. In nearly all cases, such evidence (e.g., row-wise tree orientation, log forwarding trails or landings, rectangular areas with lower crown density in a mature forest matrix) was present, even for harvests ~10 years prior. Where evidence of apparent harvesting was lacking, the location was assigned back to the “natural forest” group. We used a similar process to stratify wetlands into a “natural wetland” group, for which soil observations revealed no Ap horizon, and a “wetland restoration” group, which was comprised of locations with Ap horizons and wetland NLCD classes, indicating a wetland with a past history of cultivation. Altogether, we used these categorizations of land use or management, based on a combination of remote sensing and soil description, to arrive at statistical comparisons of SOC stocks (e.g., for harvested vs. natural forest; restored wetland vs. natural wetland) similar to those tested directly with meta-analysis.

We filled gaps in the horizon-level data for soil observations (e.g., C concentrations and Db values) based upon methods described in Nave et al. (2018). Briefly, we used % organic C as the preferred variable for representing soil organic contents, computed % organic C from % total and % inorganic C when only those data were available, and assumed % total C to represent % organic C when only % total C was measured. For Db, we preferentially used the fine earth Db (< 2mm size fraction), used the whole soil Db as a second option, and used predictions generated by USDA-NRCS (Sequeira et al., 2014) as estimates if no measurements were available. We calculated SOC storage for each horizon as the product of its C concentration, Db, and thickness. For analyses, we used as response variables either the SOC stock of the O horizon, the A horizon (uppermost 10 cm), the full mineral soil profile (sum of all

mineral horizons), or the percentage of the mineral soil profile total held in the uppermost 10 cm. Overall, our final observational datasets for analysis were comprised of 4953 individual soil horizons from 849 profiles.

As noted in Section 2.3, we also used the soil observational dataset to derive %C – Db prediction equations for those meta-analysis studies missing Db values. To accomplish this, we derived separate Db prediction equations for published studies reporting as either A, AE, or AB horizons vs. B or C horizons. We created prediction equations for these two groups using horizons from the observational dataset that were designated as either A, AE, or AB horizons ($n = 417$) vs. B or BC horizons ($n = 1225$). Prediction equations were simple linear regressions relating % organic C to fine earth Db, both with $p < 0.001$ and r^2 values of 0.44 and 0.25 for A and B horizons, respectively.

Many of the variables in the observational dataset were strongly right-skewed; we normalized these using $\ln$ transformations before running statistical tests. Tests included: (1) one-way ANOVAs (with Fisher’s Least Significant Difference for pairwise group comparisons) and simple linear regressions to test for effects of single predictor variables upon SOC stocks, and (2) best subsets regressions to identify and select multi-variate models consisting of continuous and categorical predictors (the latter as reference-coded dummy variables) on SOC stocks. We elected to perform this series of analyses—univariate followed by multivariate—for several reasons. First, running repeated univariate tests on all available observational data mirrored the iterative analytical approach that we employed in meta-analysis and was thus the most direct way to validate its findings. However, because repetitive univariate tests cannot identify significant interactions between predictor variables, we performed multi-variate tests (best subsets regression) to simultaneously assess the impacts of multiple predictor variables in a more comprehensive fashion. We ran best subsets regression once for the combined O and A horizon observational dataset, and again for the full mineral soil profile observational dataset, in order to place the O and A horizons dominantly reported in the published literature in the context of the whole soil profile. In both of these best subsets regressions, we selected the strongest multivariate model as that with the highest adjusted R^2 and closest similarity between its C_p and number of predictor variables. We assessed multicollinearity of model terms using variance inflation factors (VIFs).

For all statistical tests, we set $p < 0.05$ as the threshold for statistical significance, and in graphical representations of results, we back-transformed means and presented variances as 95% confidence intervals.

3. Results

3.1. Overall effects

Meta-analysis of published literature and univariate analyses of soil observations (O and A horizons) indicated that specific land use and management treatments were the most significant source of variation in C storage in both datasets (Table 2; Q_B/Q_T and SS_B/SS_{TO} , respectively). Harvesting had no significant overall effect on SOC in either the meta-analysis or observational datasets (Fig. 2). On the contrary, soils affected by land use change had lower SOC than their natural counterparts according to meta-analysis (Fig. 2A); this result was corroborated by comparing restored to natural wetlands and reforestation to natural forests in the observational dataset (Fig. 2B). Fire, which exhibited significant effects by meta-analysis but could not be analyzed using the observational dataset, also caused significant decreases in C storage (Fig. 2A).

3.2. Forest harvesting

Within harvesting studies, meta-analysis of published literature indicated that O horizon C stocks decreased significantly due to harvest

Table 2

Sources of variation in treatment effects on (left side) and variation in (right side) SOC storage. Factors analogous across datasets appear in proximal rows. For each factor within each dataset, the total variation (Q_T or SS_{TO}), variation attributable to that factor (Q_B or SS_B) proportion of variation explained (Q_B/Q_T or SS_B/SS_{TO}), and number of response ratios (k) or soil observations (n) is shown. Bold rows indicate statistically significant predictor variables ($p < 0.05$). See Supplementary Table S1 for predictor variable details.

Meta-analysis of treatment effects					Observations of O and A horizon SOC storage				
Factor	Q_B	Q_T	Q_B/Q_T	k	Factor	SS_B	SS_{TO}	SS_B/SS_{TO}	n
Land use/management	7.4	54.3	14	109	Land use/management	320.0	1032.0	31	1004
Practice descriptor 1	18.2	51.3	35	106					
Practice descriptor 2	14.2	35.9	39	65					
Time group	0.9	54.3		109					
Plant functional type	0.3	54.3		109	Aboveground biomass*	43.9	249.5	18	475
ECOMAP section	11.1	54.3	21	109	ECOMAP section	137.0	1032.0	13	1004
Landform group	7.1	52.1		101	Landform group	39.4	304.7	13	363
Parent material group	4.5	54.3	8	109					
					Elevation*	135.8	1032.0	13	1004
Slope group	1.9	53.4		93	Slope*	60.5	1032.0	6	1004
					Slope aspect	1.9	984.7		998
Wetness group	5.1	54.3	9	109	Natural drainage index	29.2	537.4	5	594
					Mean annual temperature*	133.4	1032.0	13	1004
Control SOC stock*	0.3	54.3		109	Mean annual precipitation*	37.0	1032.0	4	1004
Soil taxonomic order	0.4	41.3		87	Soil taxonomic order	178.3	1010.8	18	940

* Analyzed by continuous rather than categorical models (meta-regression or simple linear regression); between-group variation refers to variation explained by the model (Q_M for meta-regression and model sum of squares for simple linear regression).

(-25%, CI: -39% to -3%), while A horizons showed no significant change. In terms of soil taxonomy, Inceptisols lost SOC (-19%, CI: -27% to -10%) and Ultisols showed no harvest impacts. The landform on which harvesting occurred was the strongest predictor of variation in harvest impacts on SOC, in terms of its p and Q_B/Q_T values (Table 3). Differences between soil horizons (nearly all of which were O or A horizons) and soil orders were also significant sources of variation. Parent material group and ecoregional section were significant predictors, but were 1:1 confounded because harvesting studies were limited to two ecoregional sections with co-varying parent materials (Allegheny Mountains and Blue Ridge Mountains). Because harvesting meta-analysis studies were limited to these two ecoregions, and there was a significant difference in SOC stocks between reforestation and natural forests in the observational dataset (Fig. 2B), we conducted further harvesting analyses in the observational dataset using soils from only these two ecoregions and only those showing no evidence of past cultivation (i.e., the natural forest and harvested forest groups).

The published literature vs. the observational dataset differed

widely in SOC stocks for O (Fig. 3A), but not A horizons (Fig. 3B). O horizons were reported for 62% of the natural forests and 33% of the harvested forests (data not shown) in the observational dataset, and their C stocks were an order of magnitude greater than those reported in published literature. The A horizons in the observations dataset showed wider CIs for C storage than in the published studies, but there was no directional bias between the two datasets. Both datasets indicated that sloping landforms had significantly greater A horizon SOC stocks than one or both of the other two landforms (Fig. 3B). Furthermore, the observations dataset had sufficient replication to probe the landform dependence of A horizon SOC storage in the two dominant soil orders of these ecoregions, and indicated that sloping landforms held significantly more SOC in the A horizon in both orders.

In addition to possessing significantly greater A horizon SOC stocks, sloping landforms showed significant decreases in SOC storage due to harvesting, whether assessed by horizon (Fig. 4A) or by soil order (Fig. 4B).

Two tendencies with potentially important implications emerged

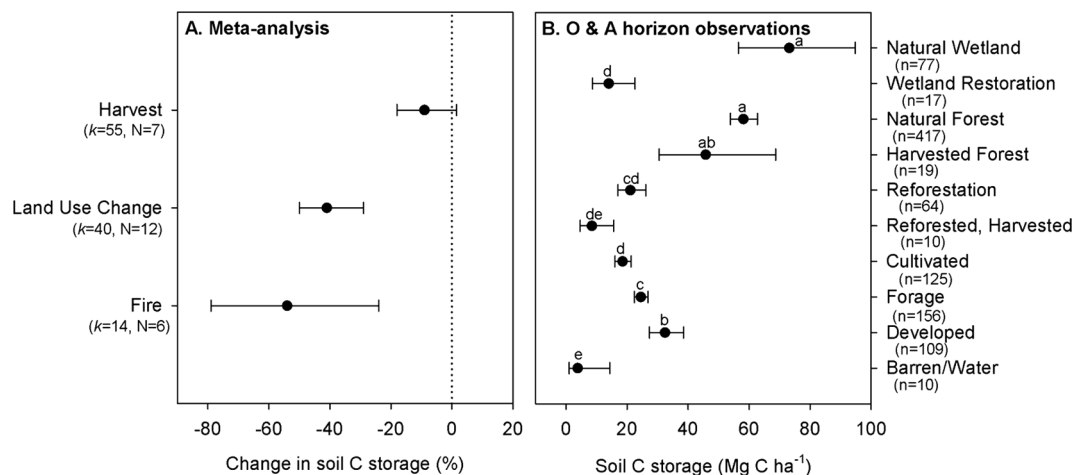


Fig. 2. Effects of land use and management on SOC storage, according to meta-analysis of published literature (panel A) and ANOVA tests of O and A horizon observations (panel B). Points and bars are means and 95% confidence intervals (CIs), respectively. In panel A, groups with CIs spanning the dotted reference line are not significantly different from 0% change, and the number of response ratios (k) and number of papers from which the response ratios were calculated (N) are in parentheses. In panel B, lowercase letters denote significantly different means (Fisher's LSD), and the number of soil observations (n) is in parentheses for each group.

Table 3

Between-group (Q_B) and total (Q_T) heterogeneity, p values, and sample sizes (k) for the predictor variables tested in the meta-analysis of SOC responses to harvesting. Factors defining groups with significantly different effect sizes are highlighted in bold.

Factor	Q_B	Q_T	p	k
Control SOC storage*	0.071	8.302	0.247	55
Practice descriptor: harvest type	0.008	7.643	0.821	53
Practice descriptor: removal intensity	0.013	6.244	0.728	51
Practice descriptor: post-harvest site preparation	0.39	3.632	0.201	55
Time category	0.262	8.302	0.435	55
Plant functional type	0.252	8.302	0.213	55
ECOMAP Section	0.572	7.644	0.047	54
Landform	2.455	7.644	0.001	54
Physiographic wetness	0.158	8.302	0.597	55
Slope group	0.536	8.293	0.191	54
Parent material group	0.572	7.072	0.048	54
Soil taxonomic order	1.109	8.302	0.027	55
Soil horizon	1.455	8.302	0.022	55
Texture class	0.875	7.632	0.112	53

* Tested with continuous meta-analysis; Q_B refers to Q_M .

during meta-analysis (Fig. 5), although they were not statistically significant predictors of variation in harvest SOC impacts in terms of Q_B/Q_T (Table 3). First, O horizon SOC declines appeared to be slightly more consistent when harvest residues were removed from harvested sites than when retained on-site; this tendency was stronger for A horizons, which lost SOC with residue removal but not when only logs were removed (Fig. 5A). Second, post-harvest site preparation and stand re-establishment practices appeared to make the difference between loss and no net change in SOC following harvest. That is, replanting maintained SOC stocks, whereas naturally regenerated stands lost SOC (Fig. 5B). Broadcast burning (with replanting) was associated with potentially large reductions in SOC.

3.3. Land use change: Sources of variation

Publications available for meta-analysis of land use change impacts on SOC storage varied widely in experimental aspects such as timescale and sampling depth. This, combined with limited study replication and confounded relationships between predictor variables and specific types of land use change, precluded a formal Q_B analysis to probe sources of variation. Nonetheless, data were sufficient to quantify SOC impacts of four generalized land use changes. Two of these land use

comparisons showed no significant differences (forest vs. developed green space, woody wetland vs. agriculture), while two showed significant differences. Specifically, young forests on former agricultural lands had SOC stocks that were 24% lower (95% CI: –35% to –17%) than mature or old growth forests, and wetlands restored on formerly cultivated lands held 67% less SOC (95% CI: –72% to –61%) than natural woody wetlands.

3.3.1. Land use change: Reforestation

The observational dataset contextualized meta-analytic land use impacts on SOC by revealing two significant patterns: (1) an overall positive relationship between aboveground biomass and SOC; (2) a consistent land use dependence of SOC across most of the ecoregions and soil orders of the study region. O horizons, which were reported for 62% of natural forest and 3% of reforestation sites in the observations dataset, were excluded from these analyses because of the apparent high bias of their C stocks noted in Section 3.2. In A horizons, SOC storage was generally higher for less disturbed land uses (Fig. 6A), being greatest in natural and harvested forests, intermediate for reforestation and cultivated lands, and least in reforested + harvested lands. Aboveground biomass C followed a similar pattern, being greatest for natural forests, intermediate for harvested forests and reforestation, and least on cultivated lands (Fig. 6B). Biomass estimates were sparsely available for the reforested + harvested group. Across the five land uses and 221 locations possessing both variables, woody biomass and A horizon C stocks were positively related (Fig. S1; regression, $p < 0.001$), with aboveground biomass C explaining 14% of the variation in A horizon SOC storage.

Significant differences in A horizon SOC stocks along the reforestation disturbance gradient were similar within soil orders as across the overall observational dataset (Fig. S2). In each of the 4 orders with sufficient data, natural forests had significantly higher SOC stocks than cultivated soils. Soils under reforestation had similar A horizon C stocks to cultivated observations in Alfisols, Inceptisols, and Ultisols, but significantly greater SOC (similar to natural forest) in Entisols. In Ultisols, where the reforested + harvested group was best represented, this most repeatedly disturbed land use had significantly lower A horizon SOC stocks than natural forests and reforestation. Ecoregional differences in A horizon C storage along the reforestation disturbance gradient (Fig. S3) were complementary to the soil taxonomic framework. Natural forests had significantly more SOC than cultivated soils in 4 of 6 ecoregional sections; one section did not have enough data for this comparison and another showed no significant difference between land

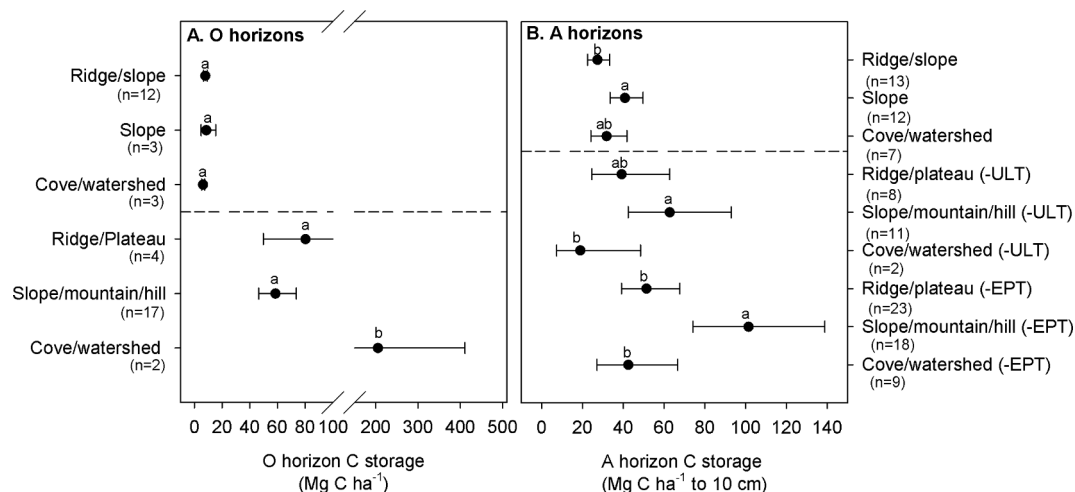


Fig. 3. C stocks for O (panel A) and A horizons (panel B, to 10 cm depth) for published studies (above dotted line in each panel) and from the observational dataset (below dotted line). Points are means, bars are 95% CIs, and lowercase letters denote significantly different groups (Fisher's LSD). Sample sizes are reported for generalized landform groups, which are arrayed in descending topographic position for the published and observational datasets, respectively. Parenthetical suffixes for A horizon observations denote Ultisols (ULT) and Inceptisols (EPT).

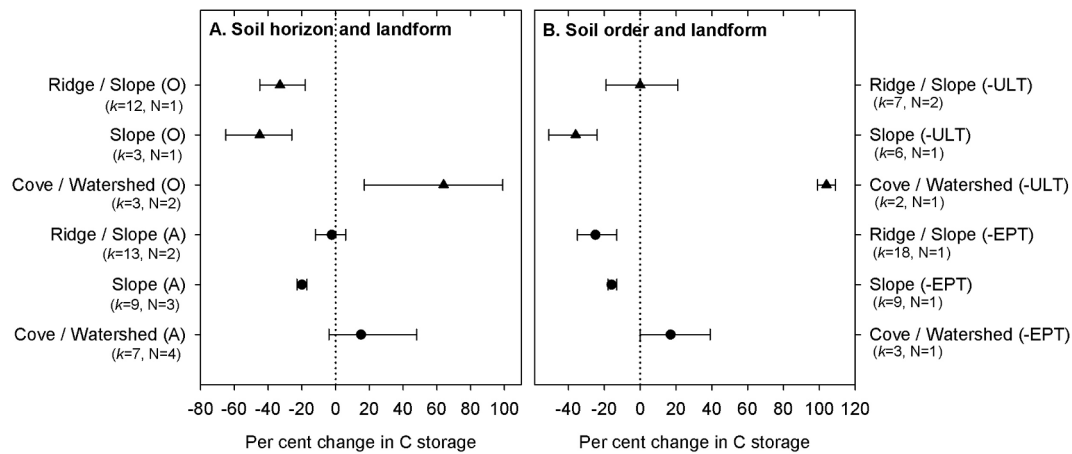


Fig. 4. Meta-analytic changes in SOC stocks due to harvesting, by landform and soil horizon (panel A) and by landform and soil order (panel B). Points are means with bootstrapped 95% confidence intervals (CIs); the number of response ratios (k) and the number of papers from which those response ratios were calculated (N) are in parentheses. Groups with CIs overlapping the dotted reference line are not significantly different from 0% change.

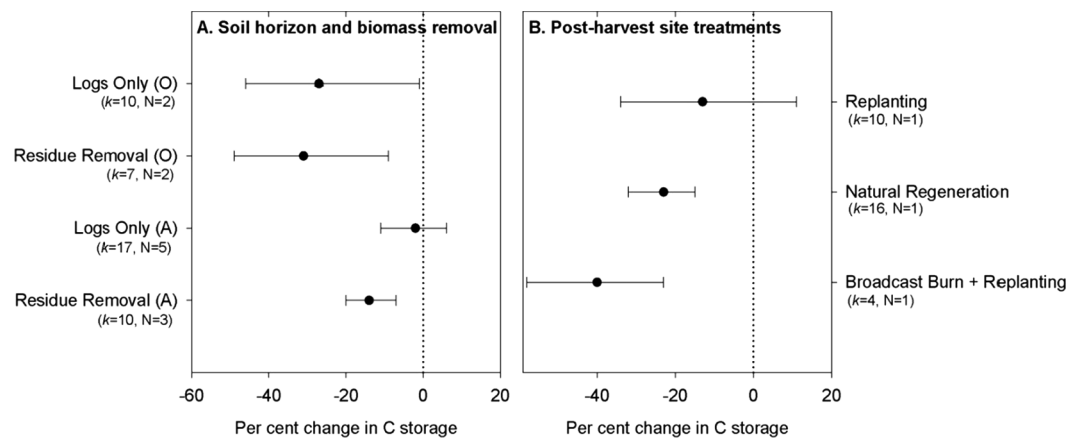


Fig. 5. Effects of harvest (panel A) and post-harvest practices (panel B) on SOC stocks, assessed by meta-analysis of published literature. Points are means with bootstrapped 95% confidence intervals (CIs); the number of response ratios (k) and the number of papers from which those response ratios were calculated (N) are in parentheses. Groups with CIs overlapping the dotted reference line are not significantly different from 0% change.

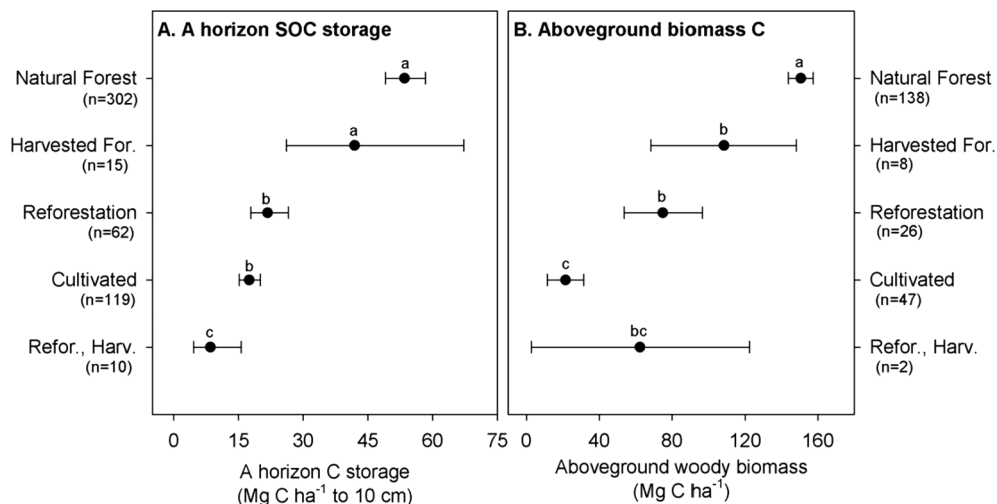


Fig. 6. C storage in the A horizon (panel A) and aboveground biomass (panel B) for locations in the observations dataset that comprise a cultivation to forest disturbance gradient. Plots show means, 95% CIs, sample sizes, and significant differences (Fisher's LSD) between groups.

uses. Harvested vs. natural forests did not differ in any ecoregion. A horizon SOC for reforestation was either intermediate between natural forest and cultivation (Northern Appalachian Piedmont), not

distinguishable from cultivation (Northern Ridge and Valley, Northern Atlantic Coastal Plain, Blue Ridge Mountains), or significantly less than cultivation (Mid-Atlantic Coastal Plains and Flatwoods).

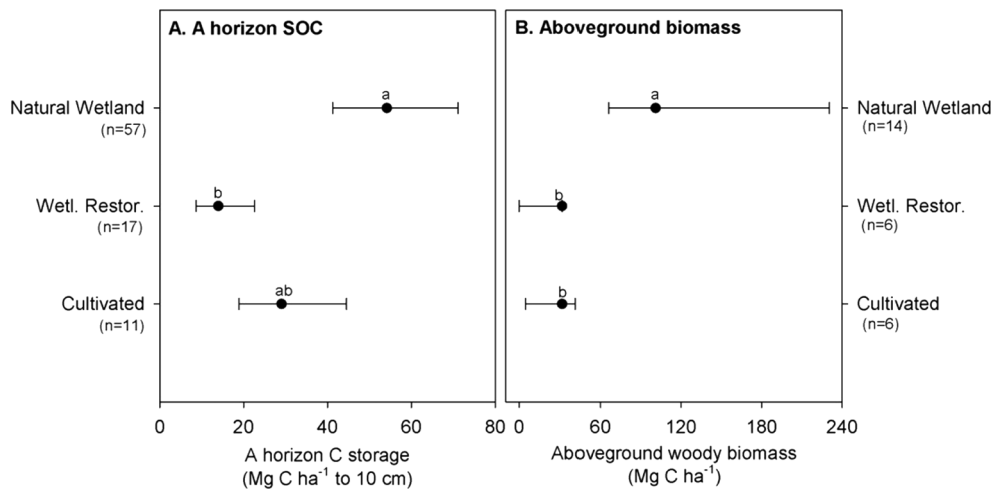


Fig. 7. C storage in the A horizon (panel A) and aboveground biomass (panel B) for locations in the observations dataset that comprise a cultivation to natural wetland disturbance gradient. Plots show means, 95% CIs, sample sizes, and significant differences (Fisher's LSD) between groups.

3.3.2. Land use change: Wetland restoration

Observational data for assessing C impacts of wetland restoration were more limited than those available for reforestation. This limitation was due to: (1) a generally limited number of observations from wetlands, and (2) a drainage criterion that we imposed on cultivated soils for this phase of the analysis. Specifically, we only included cultivated soils with poor drainage classes (SPD, PD, VPD) in these tests, in order to make a more direct comparison of wetlands to cultivated soils that possessed field-observed drainage classes consistent with wetland hydrology. Where available, data suggested that A horizon SOC stocks were significantly higher in natural wetlands than in wetlands restored on formerly cultivated soils (Fig. 7A). Observations of biomass C stocks, which were even more limited, were not significantly related to A horizon C stocks, but were significantly lower for restored and cultivated wetlands (Fig. 7B). Due to data limitations, it was not possible to assess variation in wetland land uses for distinct soil orders or ecoregions.

3.4. Fire effects on SOC

Fires had more variable impacts on SOC than harvest or land use change (Fig. 2A), and could only be assessed with meta-analysis of published data. In contrast to harvesting, which had no overall significant effect on SOC, but significant and explainable underlying variation, fires caused large SOC losses with variation that was largely unexplainable from the factors we analyzed. In lieu of a formal Q_B analysis, the available data suggested several tendencies with potentially important implications. First, and similar to harvesting, the impacts of fire were significant for O, but not A horizons, with only the former showing a significant decline in C storage (Fig. 8A). Second, fire types differed in the variability of their impacts on SOC, with highly variable (potentially large) losses following wildfires, and more consistent declines for the two types of prescribed burning (Fig. 8B). Within prescribed burns, surface fires (underburns) were associated with somewhat greater SOC reductions than fell-and-burn treatments (i.e., for xeric stand restoration).

3.5. Surface horizons in whole-profile context

Best subsets regression revealed key differences in the factors responsible for variation SOC storage in O and A horizons vs. full mineral soil profiles (Table 4). These multi-variate tests also verified trends indicated by meta-analysis and complementary tests of O and A horizon observations performed in earlier phases of the analysis. Most importantly, land use was the dominant source of variation in O and A

horizon SOC stocks. Land use variables were the 5 top-ranked variables and accounted for 7 of the 14 variables in the strongest predictive model for O and A horizon SOC storage. In contrast, land use only accounted for 3 of 11 variables in the strongest predictive model for full mineral soil profile SOC stocks, which were more impacted by soil taxonomy and profile depth. In terms of validating specific trends, dummy variable slope coefficients indicated significantly lower O and A horizon SOC stocks for barren/water, forage, developed, cultivated, reforestation, reforested and logged, and wetland restoration land uses than for the natural forest (reference) condition, while natural wetlands had significantly greater O and A horizon SOC. In terms of soil taxonomy, Inceptisols had significantly higher O and A horizon SOC stocks than Ultisols (which were the reference), while Entisols had significantly lower O and A horizon SOC storage. Compared to the (reference) Northern Appalachian Piedmont ecoregion, the Allegheny Mountains had higher, and the Middle Atlantic Coastal Plains and Flatwoods had lower O and A horizon SOC. In terms of continuously varying model terms, O and A horizon SOC stocks decreased with increasing topographic landform index, and with increasing mean annual temperature. Variance inflation factors indicated multicollinearity between mean annual temperature, the warmest and coolest ecoregions (Middle Atlantic Coastal Plains and Flatwoods and Allegheny Mountains, respectively), and landform indices.

Soil taxonomy, profile depth, and ecoregional factors were more important sources of variation in mineral soil profile total SOC storage, collectively comprising 7 of 11 variables in the strongest predictive model (Table 4). Compared to Ultisols, which were dummy variable coded as the reference, Inceptisols, Mollisols, and Spodosols had greater whole-profile SOC storage. Independent of soil taxonomy, deeper soil profiles held more SOC, and the Allegheny and Blue Ridge Mountain ecoregions were also associated with higher SOC stocks. Similar to the O and A horizon dataset, the Middle Atlantic Coastal Plains and Flatwoods ecoregion had lower full profile SOC storage, and higher mean annual temperatures were also associated with lower SOC stocks. In terms of land use, natural wetlands and developed lands had significantly higher profile total SOC stocks, while the barren/water land use category was significantly lower. Variance inflation factors indicated multicollinearity between mean annual temperature and the Middle Atlantic Coastal Plains and Flatwoods ecoregion.

Considering soil taxonomy in greater detail, the 6 most extensive soil orders in the study region differed in their mineral soil profile total SOC stocks, profile depths, and the proportions of profile total SOC stocks held in the uppermost 10 cm (Table 5). Inceptisols and Spodosols had large and overlapping profile total SOC stocks, similarly shallow profile depths, but very different depth distributions, with Spodosols

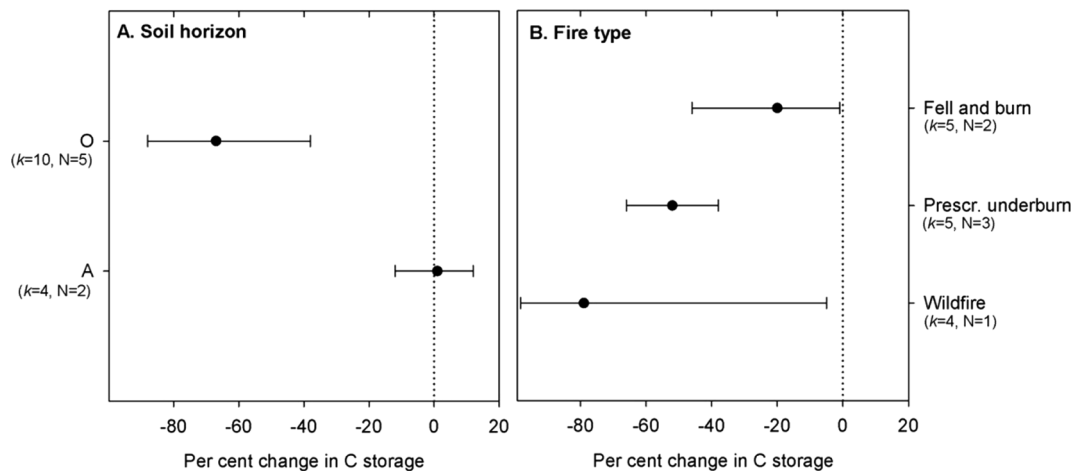


Fig. 8. Effects of fire on SOC storage for O vs A horizons (panel A) and different fire types (panel B). Points are means with bootstrapped 95% confidence intervals (CIs); the number of response ratios (k) and the number of papers from which those response ratios were calculated (N) are shown in parentheses. Groups with CIs overlapping the dotted reference line are not significantly different from 0% change.

storing a large share of C within the uppermost 10 cm of the mineral profile. Mollisol SOC stocks were not significantly different from Inceptisols and Spodosols, but showed a very deep depth distribution, with only 14% of profile total SOC in the uppermost 10 cm. Alfisols and Ultisols were similar in SOC stocks and depth distributions, but Ultisol profiles were deeper. Entisol profiles had the lowest SOC stocks, despite being among the deepest.

4. Discussion

This analysis is the first ecoregional assessment of SOC management using complementary approaches that we previously applied at biome (e.g., Nave et al., 2010) or continental (e.g., Nave et al., 2018) levels. By combining proven methods with increased data availability and hierarchical, scalable spatial frameworks (ecoregional classification and soil taxonomy), our results can begin to be applied at the more localized scales relevant to management decisions. The most important of these results affirm the first two objectives of our study. Namely, SOC stocks are responsive to management in the ecoregions of Maryland and adjacent states, and a significant share of the variation in these responses can be predicted. This is a considerable departure from arguments that SOC pools are either static through time or so variable that subtle

Table 5

Mineral soil profile total SOC stocks, depth, and percentage whole-profile SOC storage present in the uppermost 10 cm, by soil order. O horizons are excluded. Reported values are the number of profile observations, and the mean and 95% confidence interval of each parameter. Lowercase letters indicate soil orders with significantly different ln-transformed means (Fisher's LSD).

Soil order	n	Profile SOC	Profile depth	% SOC in top 10
Inceptisol	228	217 (197–239) a	97 (90–104) bc	20 (19–22) c
Mollisol	12	208 (153–283) a	116 (73–184) b	14 (10–21) d
Spodosol	15	163 (129–206) a	90 (75–109) c	31 (25–38) a
Alfisol	129	87 (81–95) b	130 (121–140) b	22 (21–24) bc
Ultisol	331	80 (75–85) b	172 (165–180) a	22 (21–24) b
Entisol	72	62 (44–89) c	159 (142–178) a	20 (17–23) c

management impacts cannot be detected with common-practice sampling schemes (Homann et al., 2008; McKinley et al., 2011; Smith et al., 2006). Our results also add needed detail to the broad concepts contained in high-level reviews (Birdsey et al., 2006; Lal, 2005; Noormets et al., 2015) by quantifying SOC changes due to management, and how these changes are mediated by tactical considerations and physical site factors. This advances the scientific basis for best management

Table 4

Best subsets regression statistics for the strongest models predicting variation in SOC stocks of O and A horizon (left side) or full mineral soil profiles (right side). Variables are ranked in descending order of (the absolute value of) their partial t statistics, and are categorized into broad factors, to aid in interpreting the most significant sources of variation. Also shown are the slope coefficients and variance inflation factors (VIF) for each model term, and the adjusted R^2 and C_p statistics for each multi-variate model.

O and A horizons ($R^2 = 0.372$, $C_p = 13.688$)					Full mineral soil profiles ($R^2 = 0.422$, $C_p = 11.16$)				
Factor	Variable	Coef.	t	VIF	Factor	Variable	Coef.	t	VIF
Land use	Constant	4.13	15.06	0.00	Taxonomy	Constant	4.85	19.23	0.00
	Cultivated	−0.78	−8.10	1.45		Inceptisol	0.81	12.17	1.54
	Barren/water	−1.97	−7.06	1.16		Natural Wetland	1.06	10.03	1.28
Land use	Forage	−0.62	−6.95	1.38	Land use	Barren/water	−1.33	−5.72	1.05
Land use	Reforestation	−0.76	−6.49	1.20	Taxonomy	Mollisol	0.97	4.84	1.03
Land use	Natural Wetland	0.67	5.55	1.52	Ecoregion	Mid. Atl. Coast Pl. & Flatwoods	−0.39	−4.70	2.01
Taxonomy	Inceptisol	0.37	5.33	1.41	Soil depth	Depth to profile bottom	0.00	2.94	1.30
Land use	Reforested, harvested	−1.29	−4.81	1.08	Land use	Developed	0.22	2.84	1.04
Ecoregion	Mid. Atl. Coast Pl. & Flatwoods	−0.49	−4.55	2.70	Climate	Mean Annual Temperature	−0.05	−2.58	3.29
Land use	Wetland Restoration	−0.94	−4.52	1.08	Ecoregion	Blue Ridge Mountains	0.16	2.11	1.51
Land use	Developed	−0.36	−3.63	1.24	Ecoregion	Allegheny Mountains	0.22	2.10	1.99
Landform	Landform Index	−0.07	−2.67	2.41	Taxonomy	Spodosol	0.37	1.94	1.14
Climate	Mean Annual Temperature	−0.05	−2.00	3.71					
Ecoregion	Allegheny Mountains	0.19	1.87	2.04					
Taxonomy	Entisol	−0.18	−1.81	1.17					

practices, many of which are based on first principles and sound reasoning, but which often lack clear connections to empirical research. However, because management tactics likely interact with site factors in ways (and at very localized scales) that we cannot address statistically, it is important not to use our results as guarantees for how much SOC is present at a specific point in a project area, nor how much that SOC stock will change after an action is carried out. Rather, our quantitative estimates of SOC stocks and changes are empirically derived guidelines that can inform land managers' expectations, and they indicate that measurements and verification should be attempted whenever possible. Careful monitoring, including measurements of all necessary soil properties (soil depths and volumes, Db, C concentrations) can make for more sensitive, place-based estimates of SOC stocks and change than the synthesis methods we have employed.

Forest harvesting is extensive across the study region, useful as a restoration tool, and includes many decision points that can influence its SOC outcomes. Our meta-analysis and observational results place these decisions into two groups: those about specific practices vs. those about where the practices are implemented. Importantly, specific harvest, site preparation, and stand re-establishment practices ("Practice descriptors" in Table 3) have less influence on SOC changes than where they are implemented. But, even subtle differences between practices (in terms of their magnitude or variability) may be important in the context of the significant site-level controls indicated by meta-analysis. For example, stakeholders may judge the potential for SOC declines with residue removal (Fig. 5) to be an acceptable risk on level sites but not on sloping landforms (Fig. 4). In such cases, the apparent SOC benefit of reforestation (Fig. 5) may be considered to mitigate SOC losses if residues are removed. As another example, if O horizon recovery is a goal in restoration of high-elevation coniferous-Spodosol ecosystems, stand-level topographic factors (Nauman et al., 2015b) may restrict harvesting (which consistently decreases O horizons) or residue removals to certain slope aspects, or justify non-harvest actions (e.g., girdling or supplemental planting) on vulnerable sites within management units. Considerations such as these become all the more important when soils are recognized as one part of ecosystem vulnerability in a region undergoing accelerating climate change. Soil moisture deficits will likely become more frequent throughout the ecoregions that we assess here (Butler et al., 2015; Butler-Leopold et al., 2018), inhibiting Spodosol formation directly and through feedbacks with the coniferous vegetation that promote their development. Independent of specific practices or ecosystem types, the large stocks and greater harvest sensitivity of A horizon SOC on slopes (Figs. 3, 4) warrant special consideration. Protection against erosion, which is one mechanism for SOC loss from slopes, is a BMP (SFI, 2015) that will only become more critical as intense precipitation continues to increase.

Reforestation—in particular, on historically cultivated lands—is widespread in the study region, and our observational results indicate that this long-term land use change intersects with forest harvesting in important ways. First, reforestation is generally well advanced in terms of forest regrowth (Fig. 6B), indicating that forests have been in recovery for sufficient time to produce substantial quantities of biomass for harvesting. However, SOC lags aboveground recovery; even as biomass C under reforestation is approaching natural forest levels, sites under reforestation have significantly lower SOC (Fig. 6A). The consequences of harvesting thus emerge from comparisons of reforested + harvested sites to natural forests, harvested forests, and lands under reforestation that have not been harvested (Fig. 6). Namely, harvesting a natural forest has no apparent SOC impact, but it significantly decreases SOC in forests harvested on formerly cultivated lands. This pattern is consistent with the compound disturbance literature (Buma et al., 2014; Hume et al., 2018). Our results for the less productive Entisols (Figs. S2, S3) suggest particular vulnerability of reforestation in these settings. When harvesting is conducted in these settings, southern pine forestry provides many examples of management options to maintain or enhance the productivity of historically

depleted soils (Fox et al., 2007; Vance, 2018). Considering the longer term, the gap between reforestation and natural forest conditions (Section 3.1, Fig. 2A), and the potential for mined land afforestation (Nave et al., 2013) suggests that forest re-establishment can play a prominent regional role in terrestrial C sequestration.

Our results for wetland restoration (i.e., on formerly cultivated lands) are similar in pattern and interpretation to our results for reforestation; that is, the greater C stocks in soil and biomass in natural vs. restored wetlands (Figs. 2, 7) indicate the C storage benefit of maintaining wetlands in their natural condition. However, the magnitude of these differences may not be as large as our statistical comparisons suggest if wetland restoration (or incidental recovery) is preferentially occurring on sites where SOC is either inherently low, or has been depleted by cultivation or restoration practices (Stolt and Rabenhorst, 1987; Fenstermacher et al., 2016). It is also important to recognize the limitations of remote sensing for identifying wetlands in either their natural or converted conditions. In this study, we were able to critically appraise the wetland assignments made by NLCD, using the natural drainage index classifications recorded by field soil surveyors. These classifications were available for 72% of geolocations in our observational dataset (Table S2). For profiles categorized as wetlands by NLCD, 64% had drainage classes of somewhat poorly drained (SPD) or wetter; forests were the next wettest land cover group, with 15% of profiles possessing a drainage classification of SPD or wetter. Other land cover groups had 8–14% of their profiles classified as SPD or wetter. In contrast, 18% and 14% of wetlands were reported as well-drained and moderately well-drained, respectively, while these classes were 57–68% and 15–23% of other land cover groups. Overall, these comparisons suggest that NLCD recognizes wetlands as such approximately two-thirds of the time, although it does not detect modest proportions of lands under other cover types (or uses) that are in fact wetlands. Given that wetlands have high SOC densities and are extensive throughout the study region, there is a clear need to increase wetland soil C data coverage (Holmquist et al., 2018). However, it may also be that in the coastal wetland systems of the Mid-Atlantic, management decisions have less impact on SOC than external forces such as inputs of water, C and nutrients from contributing watersheds, sea level rise, sedimentation, and saltwater intrusion (Bridgman et al., 2006; Butler-Leopold et al., 2018; NOE et al., 2016). Therefore, though our insights into SOC management in these systems are limited, so too is the potential for altering their large SOC stocks through management.

Our meta-analysis results suggest that fire may decrease SOC stocks quite severely (Fig. 2a), but the limited extent of fires translates to a fairly small region-wide impact. However, pest outbreaks, tree mortality, and fuel accumulation may increase the extent or severity of fires in the region as climate continues to change (Butler-Leopold et al., 2015). Further, fire is increasingly used for fuel reduction and xeric species or community restoration (Nowacki and Abrams, 2008). In light of the generally greater declines in SOC due to wildfire than prescribed fire (Nave et al., 2011), our present results can help prioritize fire use, and contextualize its impacts against management alternatives in the region. Most importantly, the large O horizon losses and high variability of SOC change after wildfires indicated by our results are concerning for a number of reasons. Due to increasing and increasingly episodic precipitation in the region (Butler et al., 2015; Butler-Leopold et al., 2018), more extensive or severe wildfires could create more extensive areas of destabilized slopes, erosion, challenges to forest regeneration, and feedbacks to soil productivity (Dey et al., 2019). More locally, proactive management of fuels or stem density through prescribed under-burning or fell-and-burn stand restoration practices may help to restore ecosystems while preventing wildfires and attendant SOC losses. The use of prescribed fire may therefore mitigate SOC losses, while preventing the need for costly and less effective reactive management (Elliott et al., 2013; Nowacki and Abrams, 2008). Lastly, when fire is used in landscape level management, consideration of the impact of past fires might constrain other actions such as harvesting on

vulnerable sites or soils recovering from compound disturbances.

The multi-variate framework of best subsets regression validated the results of meta-analyses and the complementary univariate tests of the observational data. It also allowed for more integrative interpretations, by indicating that multiple independent predictor variables simultaneously influence SOC stocks, and by placing O and A horizons in the context of whole soil profiles. In terms of validation, the importance of land use and management to SOC stocks in O and A horizons was clear from the abundance of significant relationships with land use variables. Of particular note were the findings that the slope coefficients for reforestation, cultivated, and reforested + harvested land uses (in that order) indicated significantly lower O and A horizon SOC than natural forests. The significant continuous relationship between O and A horizon SOC and the topographic landform index variable was another validating result. The negative relationship extends the interpretation of our meta-analysis and univariate tests (Figs. 3, 4) that revealed larger stocks and vulnerabilities of surface soil SOC in sloping settings, but which were limited to those observations with defined landform categories. Smaller landform index values correspond to ridge and slope landforms, while larger index values correspond to cove landforms (McNab, 1993), which in contrast to slopes, had smaller SOC stocks that were not diminished by harvesting. This suggests the potential to use similar GIS-based approaches to identify sloping landforms vulnerable to O and A horizon SOC loss in the study region.

Best subsets regression identified Inceptisols, Mollisols, and Spodosols as the orders with large whole-profile C stocks. In light of these taxonomic differences, the simultaneous, positive relationship between profile depth and SOC highlights the inferential benefits of multi-variate tests. Namely, Spodosols and Inceptisols had larger SOC stocks, despite having significantly shallower profiles than other orders (Table 5); thus the overall inference is that soil depth and taxonomy are independent (and potentially interacting) sources of variation. The implications of this inference are most relevant in mountainous ecoregions (which also had significant positive relationships with profile SOC), where these orders are most common. In mountain landscapes, soil taxonomic units may be quite large (e.g., landform-level), but even with them, localized variation (e.g., microtopography or depth to bedrock) creates localized variation in profile depth and SOC stock. Where projects are conducted in such settings, more detailed information (e.g., high-resolution DEMs or soil maps) can inform decisions intended to manage SOC on vulnerable soils.

Many of our results have implications for management—in terms of practices and where they are implemented—by virtue of significant differences between observational group means or meta-analytic effect sizes. The implications of variability in our results are at least as important. First, in any analysis, whether meta-analytic or observational, no statistical model explained more than 42% of the variation among observations. While statistically significant, this predictive capacity is not a strong basis for risk-averse management, and indicates that factors we cannot test (or in some cases even know) nonetheless influence SOC in most every case. These myriad factors may include everything from the time elapsed since disturbance(s), to microtopography, to methods execution, and interactions among them. This points to what is possibly the most important inference that can be drawn from the evident variation between sites and studies in our datasets. Namely, the predictive utility of land use and management is much diminished when moving from O and A horizons to the whole profile, where soil taxonomic and ecoregional variables provide greater predictability. This difference in predictor variables for surface horizons vs. the whole profile is essentially the basis for site-specific management, which integrates all aspects of the ecosystem (Barnes, 1996). Indeed, in our multivariate models, variables for place (ecoregion) showed multicollinearity with physiographic factors (landform index, mean annual temperature), highlighting the reality that statistical predictor variables are not truly independent, but integrative, and co-varying with place. In the context of SOC, site-specific management considers the superficial

influence of management against the bottom-up factors that define the unique soil units on a landform, landscape, or in an ecoregion, and makes tactical decisions based on the constraints imparted by the site (soil). Ultimately, because our synthesis methods cannot address all site-dependent factors, for all sites, our empirical results are best used as one input to decision making processes. In the end, considerations such as risk tolerance, institutional or stakeholder constraints, and managers' site-level knowledge are at least as important for making deliberate decisions about SOC management as the results we report here.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.foreco.2019.05.072>.

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